

Key-Value Insertion

Topic.

Implicit-key data structures, treap, array insertion with shifts.

Statement.

You need to maintain a data structure for integer key values.

The structure is an infinite array A whose cells are numbered starting from 1.

Initially, all cells are empty.

The only operation is

$$\text{Insert}(L, K),$$

where L is a position in the array and K is a positive integer key value.

The operation is performed as follows.

If $A[L]$ is empty, assign

$$A[L] := K.$$

Otherwise, perform

$$\text{Insert}(L + 1, A[L]),$$

and then assign

$$A[L] := K.$$

Given a sequence of integers

$$L_1, L_2, \dots, L_N,$$

output the contents of the array after performing

$$\text{Insert}(L_1, 1),$$

$$\text{Insert}(L_2, 2),$$

...

$\text{Insert}(L_N, N)$.

Input.

The first line contains two integers

N, M ,

where N is the number of insert operations and M is the maximum position that can be used in an insert operation.

The second line contains N integers

L_i .

Output.

Print W , the last non-empty position in the array.

Then print

$A[1], A[2], \dots, A[W]$.

For empty cells, print 0.

Implementation.

The implementation is provided separately in the file

`solution.cpp`.